

Dataset 1: Chest X-ray

- Type: X-ray (chest radiographs)
- Number of images: 112,120 images from 30,805 patients
- Classes/labels: 14 disease labels (e.g., Atelectasis, Cardiomegaly, Effusion, Pneumonia, Pneumothorax, etc.) multi-label, extracted via NLP from radiology reports
- Dataset imbalance: Yes "No Finding" dominates (60%), while diseases like Hernia have only a few hundred images
- Challenges:
 - Labels are NLP-derived, not radiologist-verified → noisy annotations
 - Multi-label nature makes classification harder
 - Class imbalance affects rare disease detection

Summary: Chest X-ray is a large-scale chest X-ray dataset useful for multi-label disease classification research. Its size is a major strength, but the NLP-generated labels and heavy class imbalance make it noisy and challenging for training reliable models.

Dataset 2: Brain Tumor MRI Dataset (Kaggle)

- Type: MRI (brain scans)
- Number of images: 7,023 images
- Classes/labels: 4 classes Glioma, Meningioma, Pituitary tumor, No tumor
- Dataset imbalance: Mild-moderate "No tumor" and "Pituitary" classes have slightly more images than Glioma/Meningioma, but not extreme
- Challenges:
 - Images sourced from multiple datasets → variation in scan quality, contrast, and orientation
 - Small dataset size relative to deep learning needs
 - Tumor boundaries can be subtle, making annotation/classification harder

Summary: The Brain Tumor MRI Dataset is a moderately sized collection used for tumor type classification. It's more balanced than typical medical datasets, but its mixed sourcing introduces scan-quality inconsistencies, and its smaller size limits deep model training.